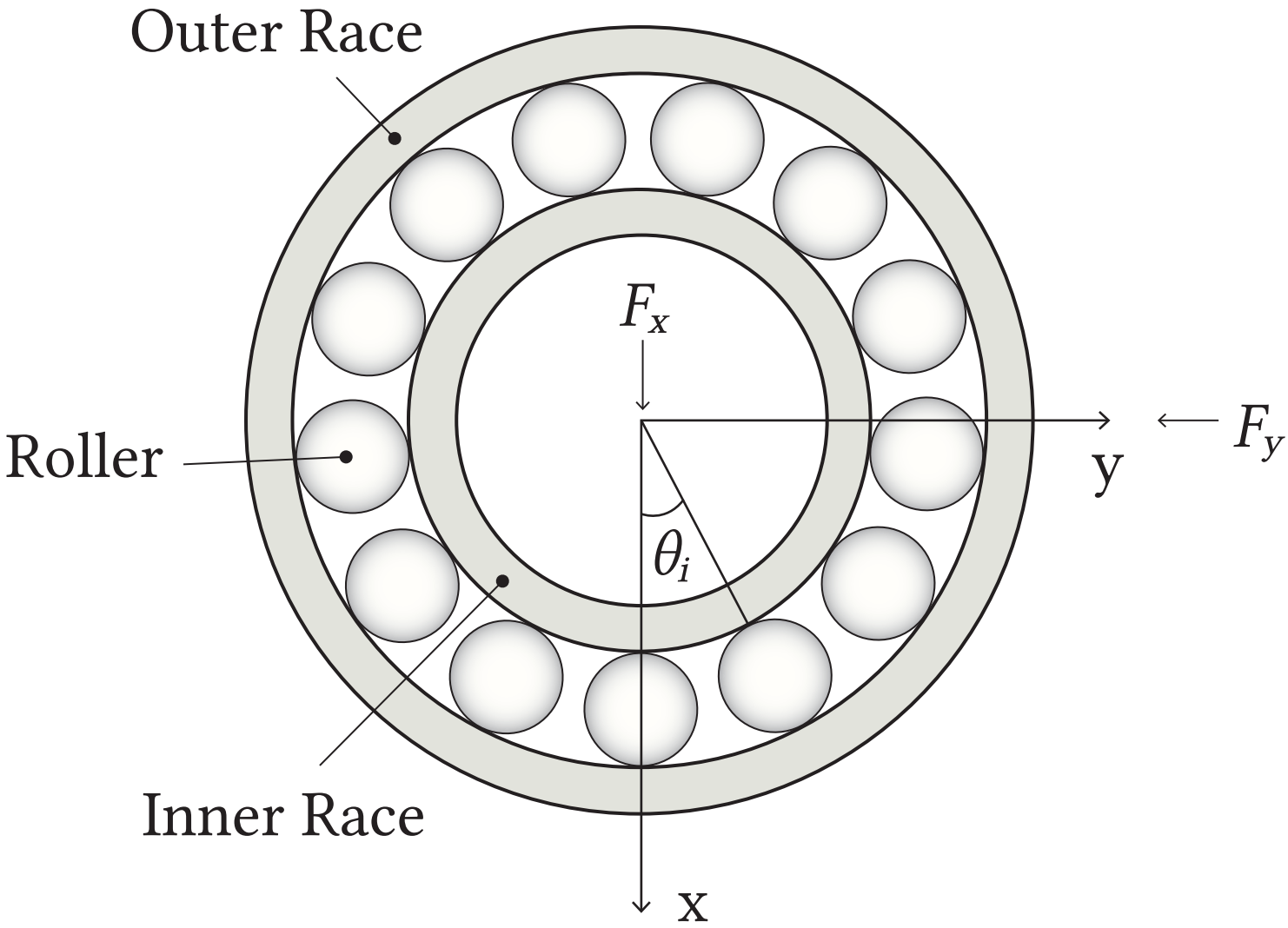
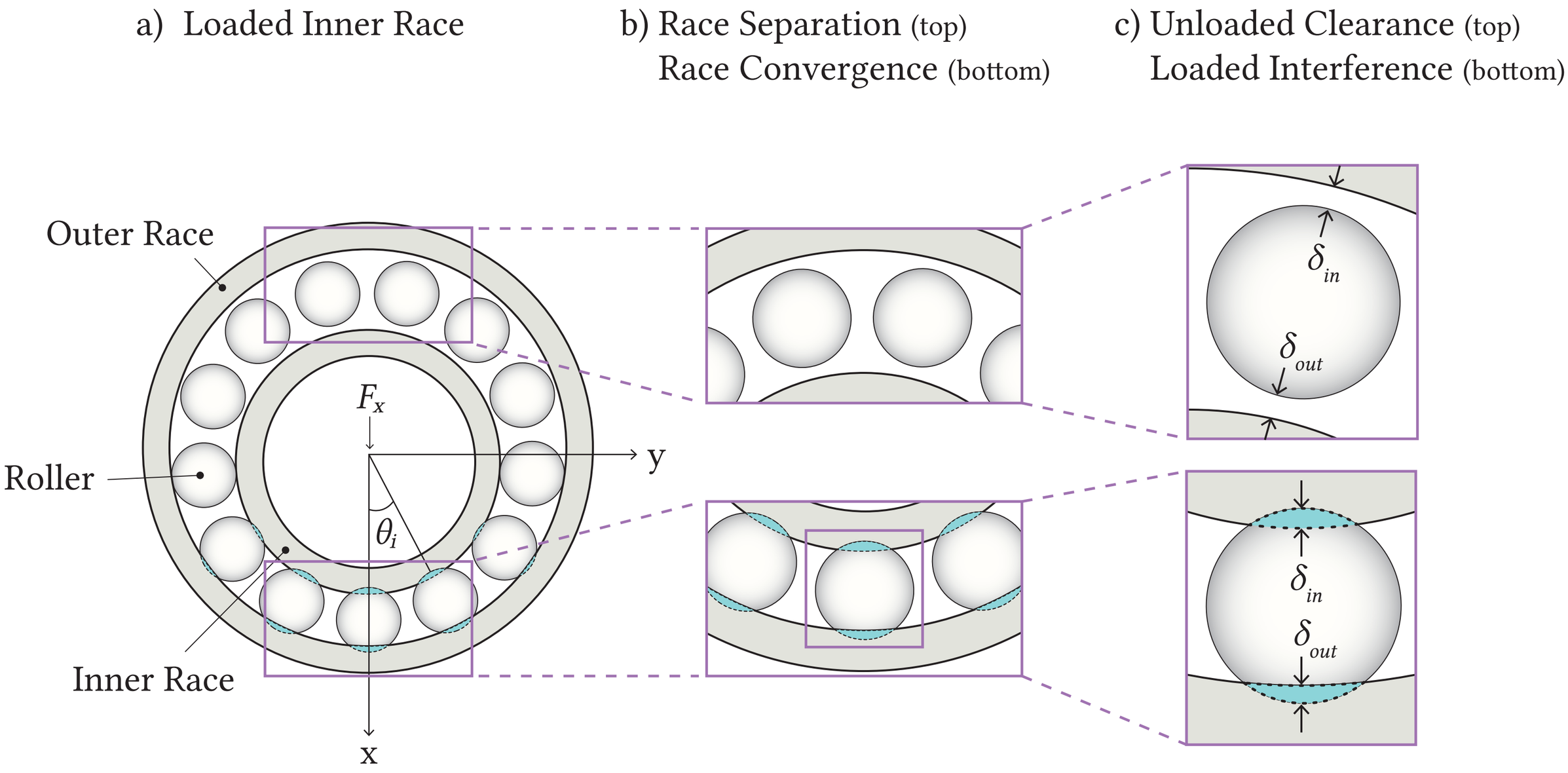
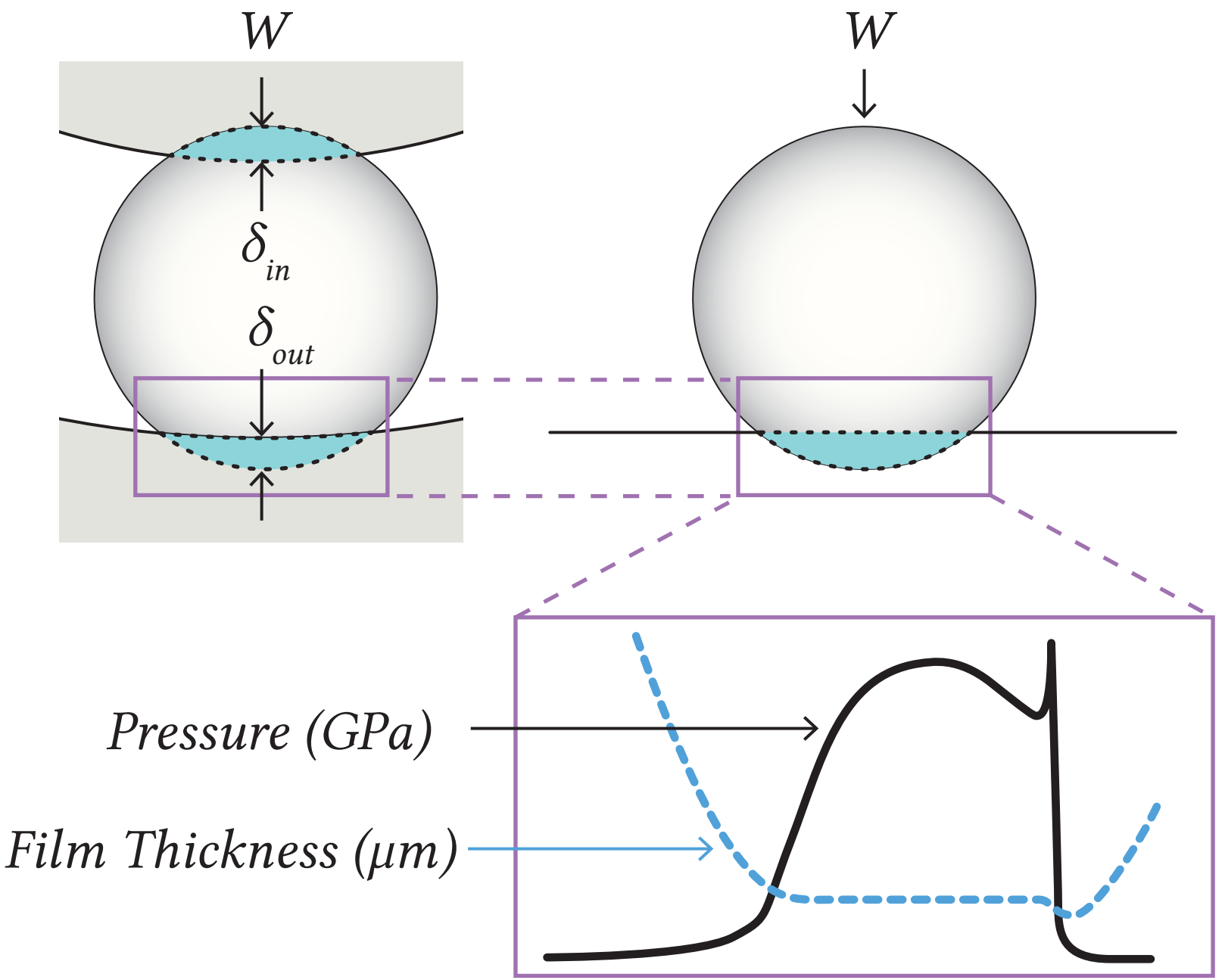
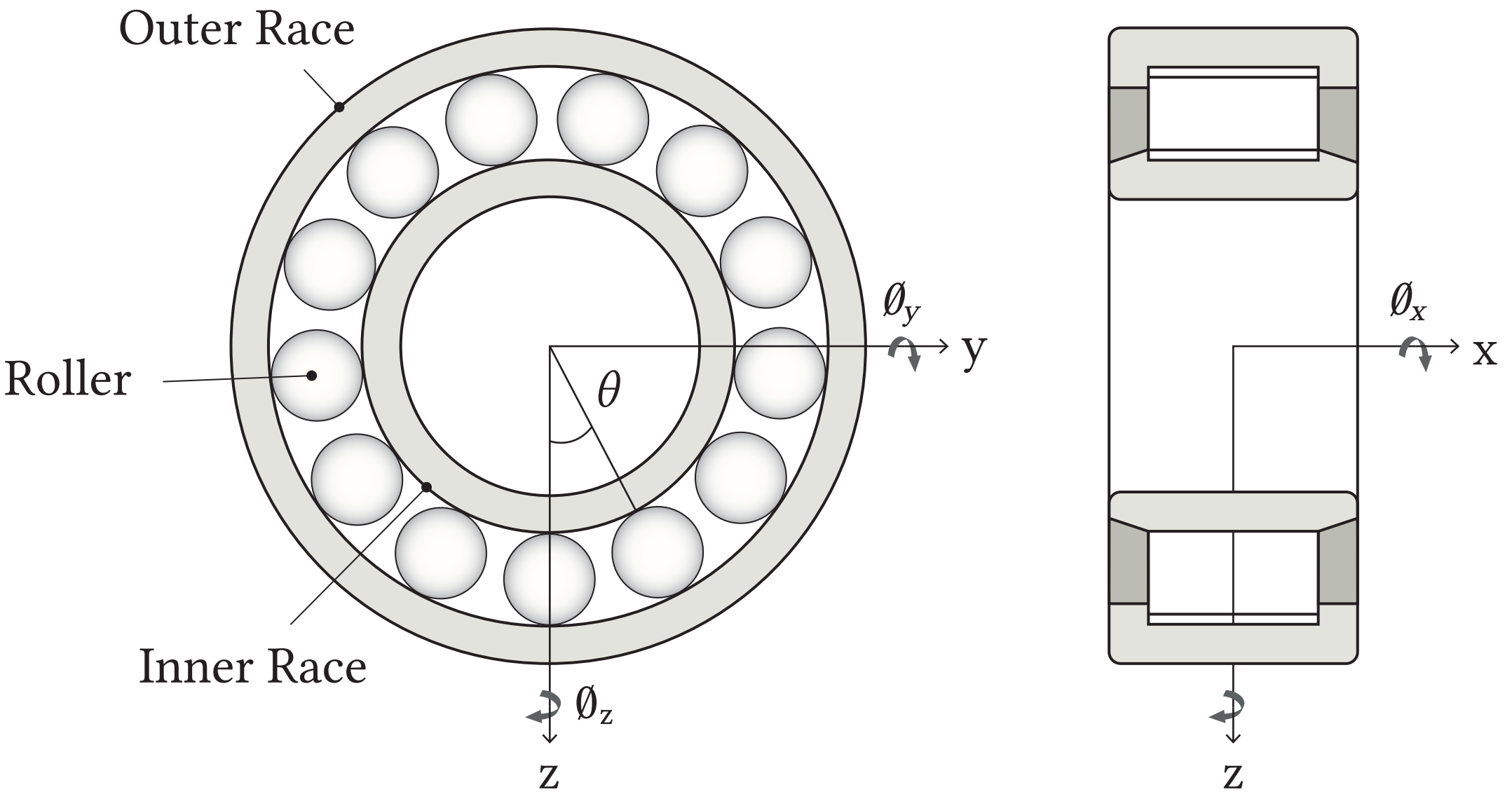
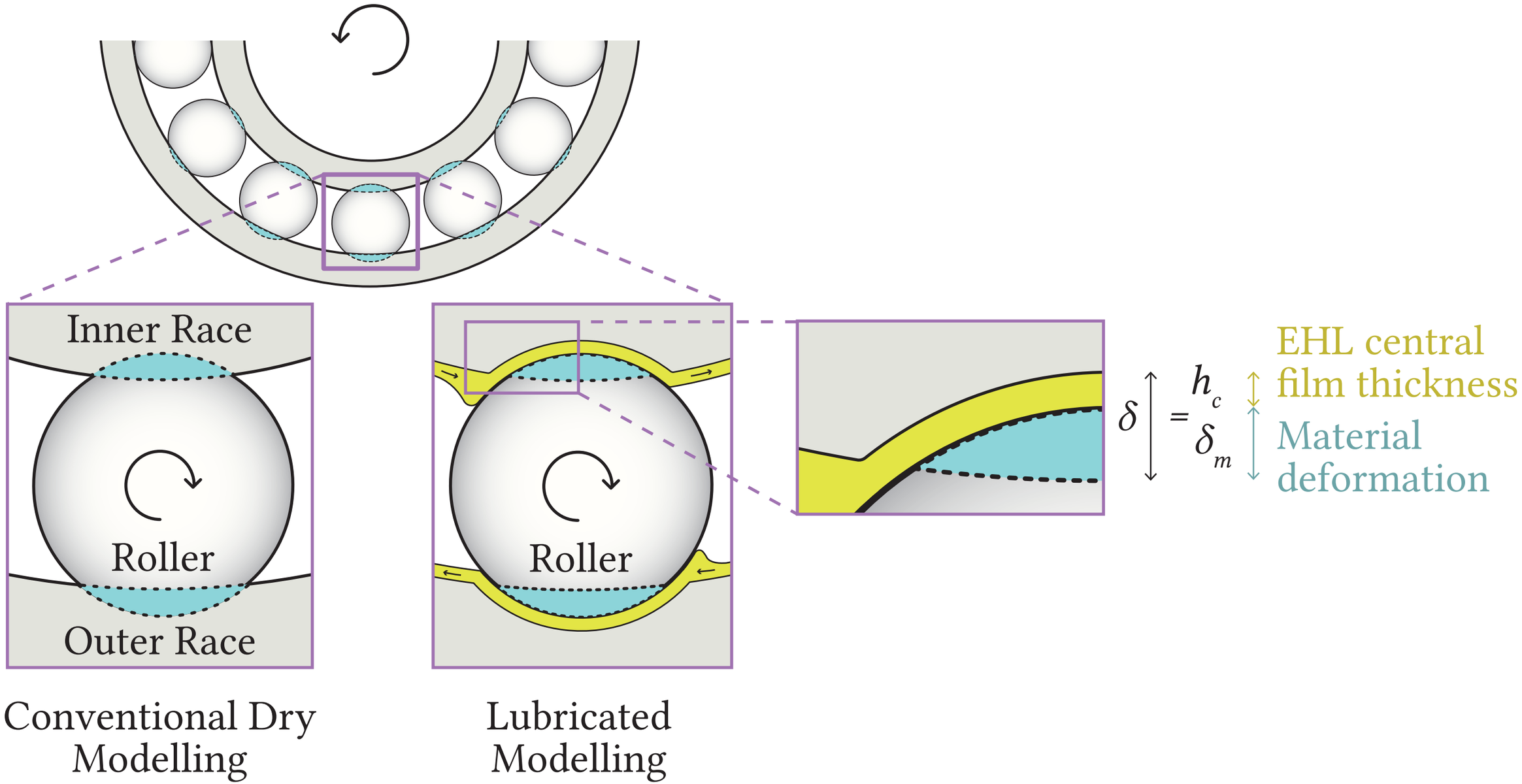
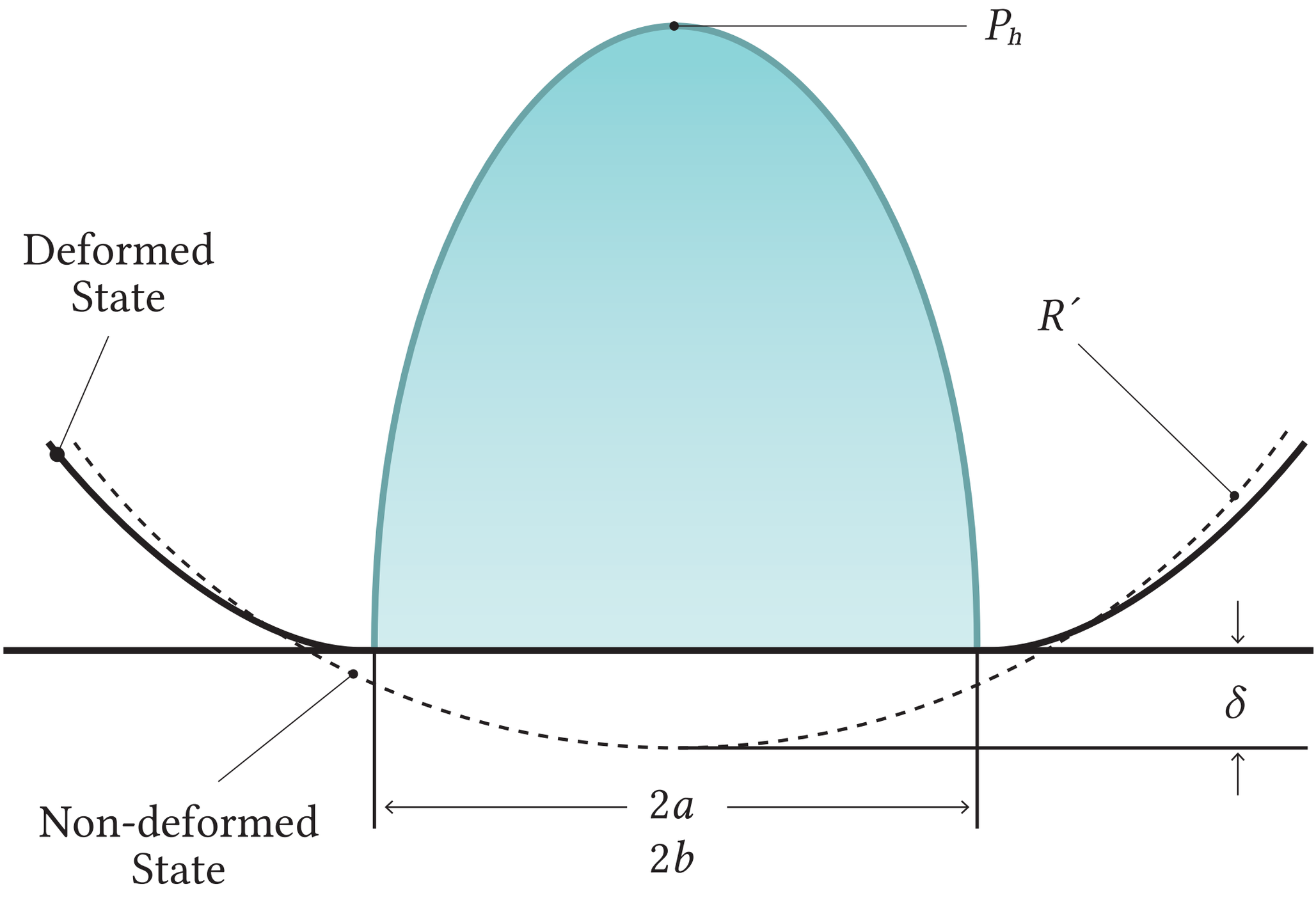

Outer Race
Roller
Inner Race
F_x
F_y
θ_i
y
x

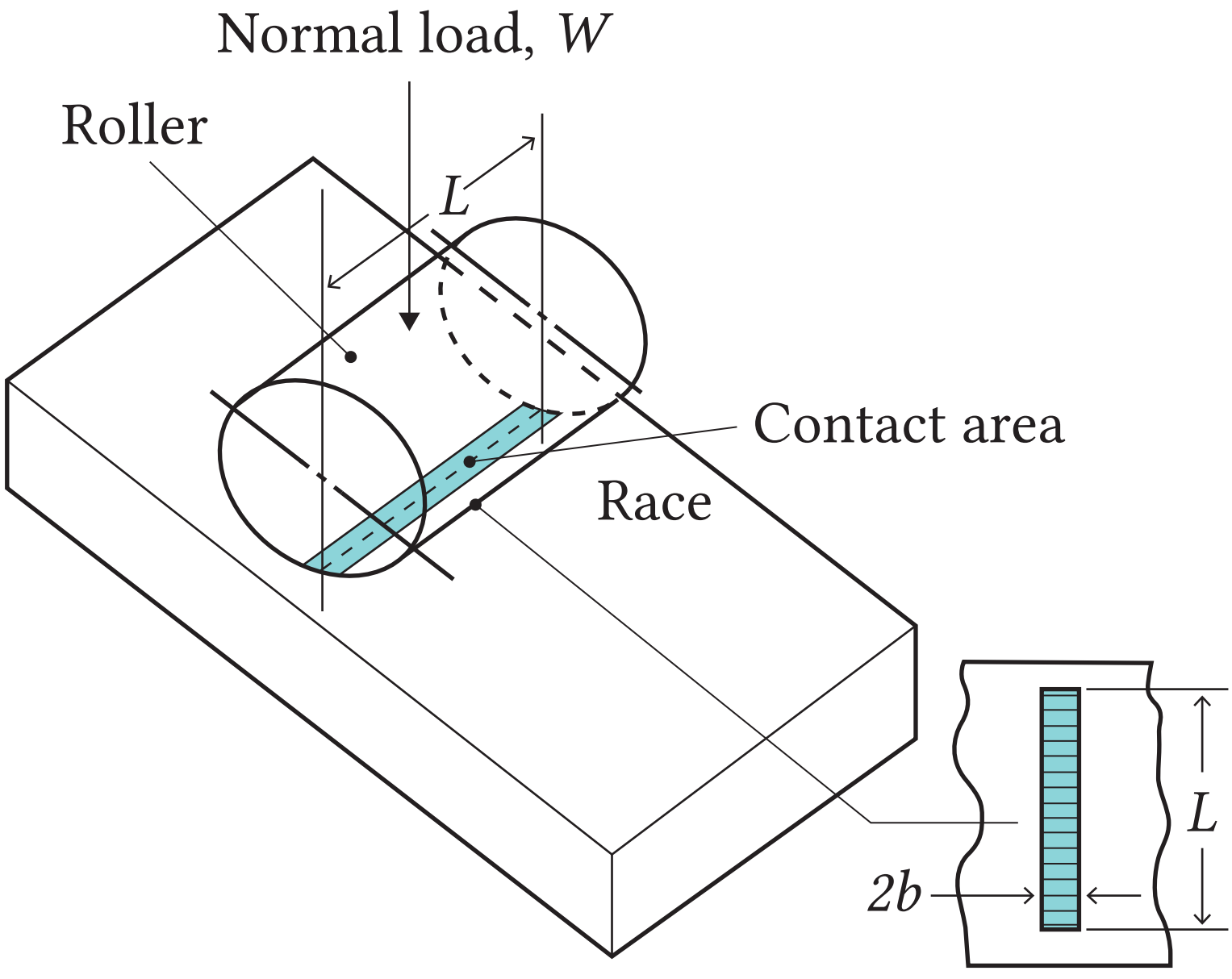

a) Loaded Inner Race
b) Race Separation (top)
Race Convergence (bottom)
c) Unloaded Clearance (top)
Loaded Interference (bottom)
Outer Race
Roller
Inner Race
F_x
θ_i
y
x
δ_{in}
δ_{out}
δ_{in}
δ_{out}

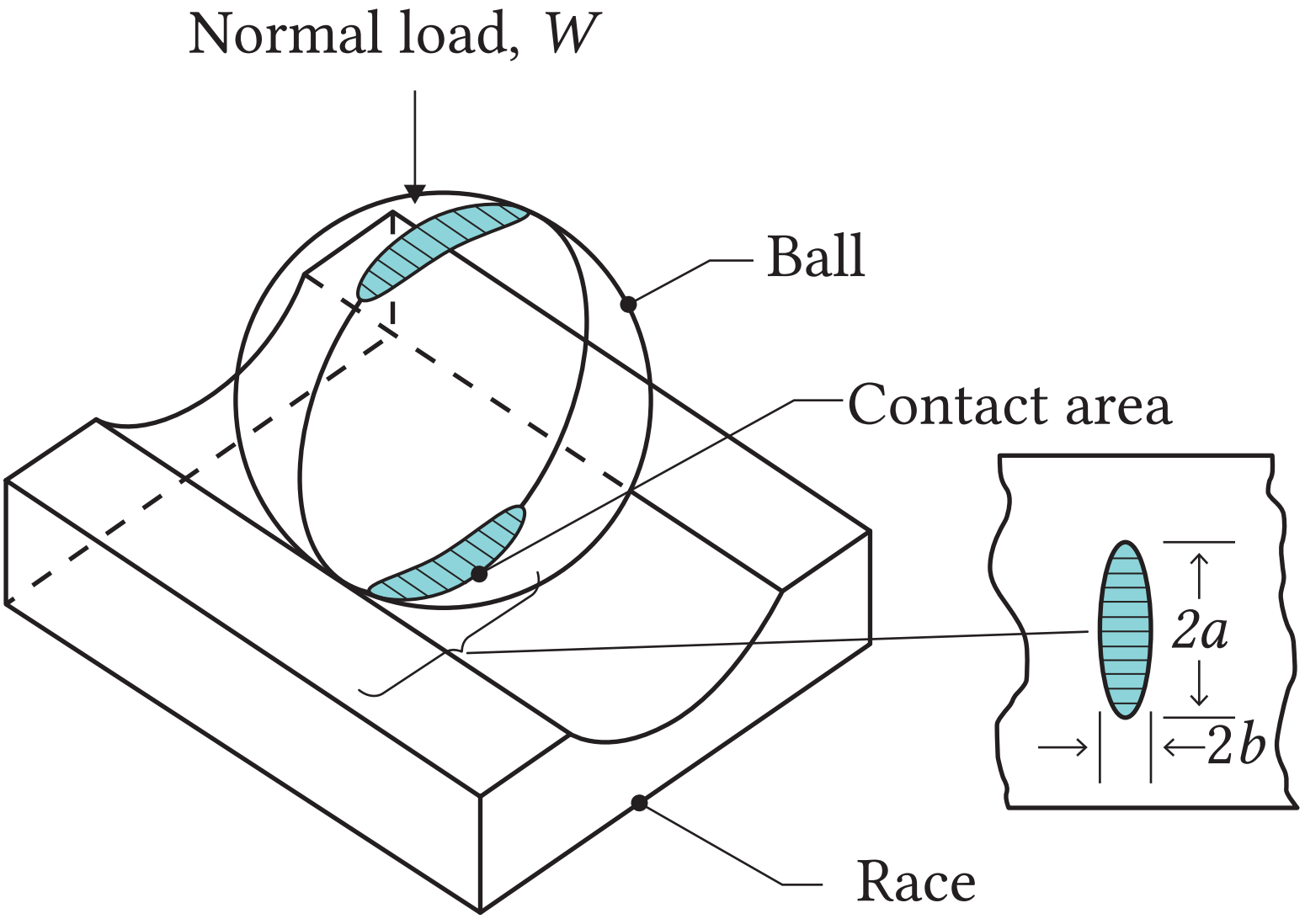

W
W
δ_{in}
δ_{out}
Pressure (GPa)
Film Thickness (μm)

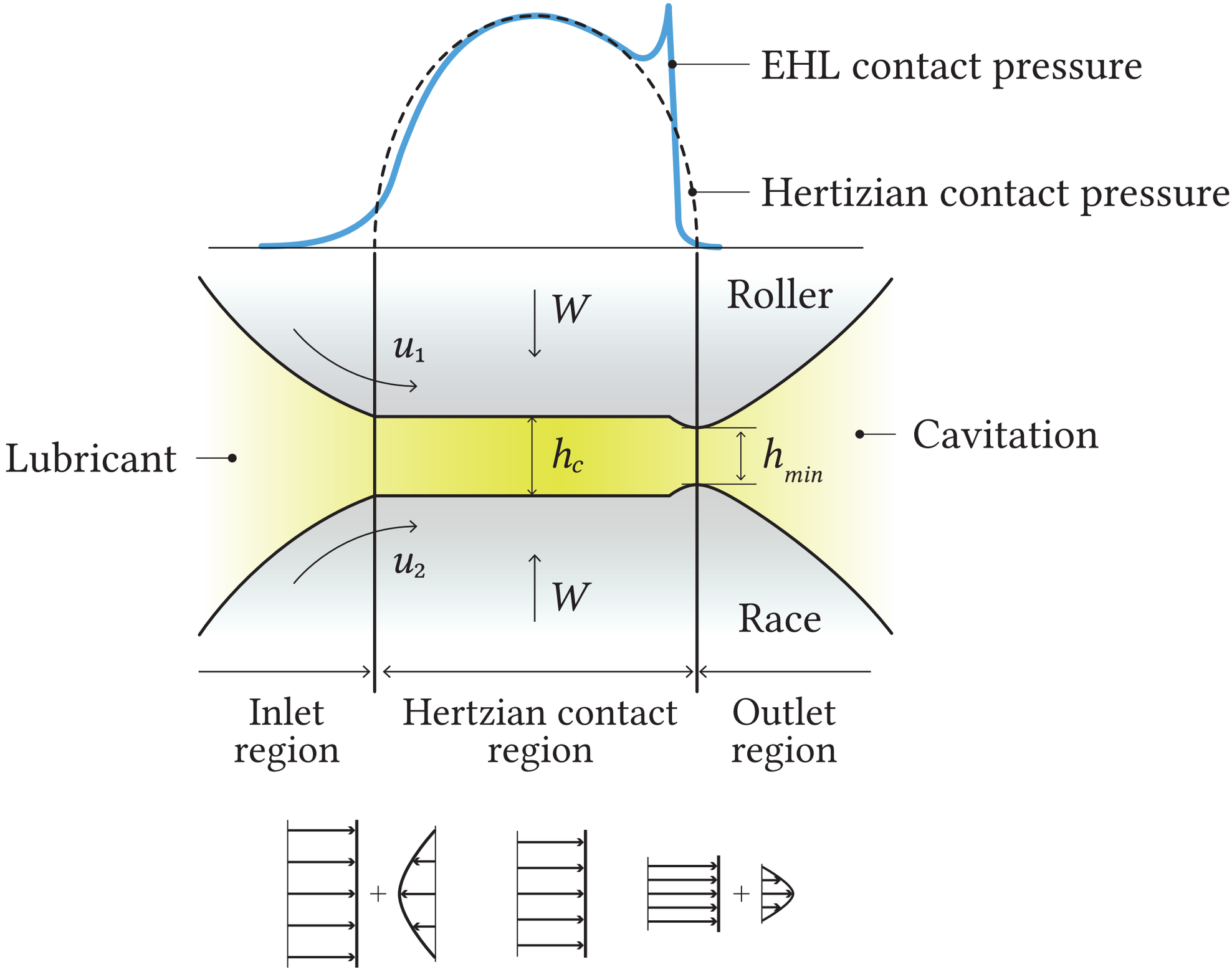

Outer Race
Roller
Inner Race
θ
Ø_y
y
Ø_z
z
Ø_x
x
z


Inner Race
Roller
Outer Race
Conventional Dry Modelling
Roller
Lubricated Modelling
$\delta = h_c + \delta_m$
EHL central film thickness
Material deformation


P_h
Deformed State
R'
δ
Non-deformed State
$2a$
$2b$


Normal load, W
Roller
L
Contact area
Race
L
2b


Normal load, W
Ball
Contact area
$2a$
$2b$
Race


EHL contact pressure
Hertizian contact pressure
Roller
W
u_1
Lubricant
h_c
h_{min}
Cavitation
u_2
W
Race
Inlet region
Hertzian contact region
Outlet region
+
+